Praneeth Reddy Gandra

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SUMMARY

Passionate about **machine learning**, **generative AI**, and **AI agents**. I train and fine-tune models, build **LLM-powered agents** and RAG pipelines, and develop end-to-end ML solutions for real-world problems.

EDUCATION

IIIT Tiruchirapalli 2023 – 2027
B.Tech, Computer Science and Engineering Trichy, Tamil Nadu

- **CGPA**: 8.96 / 10.0 | **Semester 5 GPA**: 9.38 / 10.0

EXPERIENCE

Quant Singularity | *AI/ML Intern* Jun 2026 – Present

- Built a production forecast aggregation layer for multi-strategy trading signals using weighted ensemble voting, conflict resolution, and regime-aware routing.
- Integrated production **XGBoost** and **LightGBM** models with an **MLflow**-based serving pipeline, enabling low-latency online inference with offline fallback.
- Defined feature-schema contracts, offline replay pipelines, and inference APIs over **structured market data**; validated with **368** automated pytest cases while meeting a **p99 latency under 50 ms**.

Magnivo AI | *Agentic AI Intern* May 2026 – Jun 2026

- Shipped a production **multi-agent GTM automation platform** built on **specialized LLM agents** for lead discovery, enrichment, personalized outreach, and campaign execution.
- Engineered the **prompt engineering** and **tool-calling** layer behind each agent, adding structured outputs, schema validation, and retry/fallback handling to keep multi-step runs reliable on **unstructured web data**.
- Orchestrated agent workflows with shared state and integrations including **SerpAPI**, **Hunter**, Gmail, and Instantly; validated production reliability with **152** automated tests.
- Integrated the platform with **Next.js**, **Clerk**, **Supabase**, and **Render**, enabling CRM synchronization, automated outreach campaigns, and engagement analytics.

IIT Madras | *Research Intern* Jan 2026 – Apr 2026

- Researched **additive-manufactured surface metrology**, evaluating nonlinear least-squares mean-line fitting and morphological filtering for surface roughness characterization.
- Developed an end-to-end Python evaluation pipeline using **NumPy** and **SciPy** to benchmark both methods across **19** stainless-steel optical profiler scans with standardized preprocessing.
- Found mean-line fitting achieved **3x** higher consistency than envelope filtering, reducing Sa variability from **77.5%** to **23.7%** and identifying failure cases on spatter-heavy surfaces.

PROJECTS

Study Assistant Hub | *LangChain, Agentic RAG, FAISS, Groq, FastAPI* | live demo Nov 2025 – Dec 2025

- Built an agentic study platform that generates quizzes, flashcards, planners, mind maps, and summaries from lecture PDFs using retrieval-augmented generation.
- Implemented a grounded RAG pipeline with query rewriting, retrieval, grading, and correction loops using **LangChain** and **FAISS** to improve answer reliability.
- Served the pipeline through a **FastAPI** backend with Groq-hosted inference, handling PDF parsing, chunking, and embedding of unstructured lecture material.

LLM Fine-Tuning for Financial Sentiment Analysis | *Unsloth, PEFT, QLoRA* Feb 2026 – Mar 2026

- Fine-tuned Meta's **Llama 3.2 (3B)** model using **QLoRA** and **Unsloth** for financial sentiment classification under 4-bit quantization.
- Applied **PEFT** instruction tuning on a domain-specific finance dataset to reduce memory requirements and improve training efficiency.

Swizzo | *Swiggy MCP, Claude Agent SDK, Next.js, TypeScript* Jul 2026 – Aug 2026

- Built an AI commerce concierge using **Swiggy MCP** that searches live products and restaurants, compares options, tracks orders, and assists users through conversational shopping.
- Designed a safe agent architecture where LLMs handle reasoning and recommendations while backend services control cart updates, coupons, and order placement with user approval.
- Developed the **Next.js** frontend with streaming chat, live product cards, cross-restaurant search, and real-time cart sync.

FDM 3D Printing Defect Detection | *PyTorch, Computer Vision, Transfer Learning* Jan 2026 – Feb 2026

- Trained CNN classifiers including **AlexNet**, **ResNet18**, and **InceptionV3** using ImageNet transfer learning for FDM defect detection.
- Achieved **99.45%** binary and **99.22%** multi-class accuracy across printing defects including cracking, layer shifting, stringing, and warping.

SKILLS

GenAI & LLMs: Prompt Engineering, LLM APIs (OpenAI, Claude, Gemini), Tool Calling, MCP, RAG, Agentic RAG, FAISS, Fine-Tuning (PEFT, QLoRA), Hugging Face

Agentic AI: Multi-Agent Systems, Agent Orchestration, LangChain, LangGraph, Claude Agent SDK, Vercel AI SDK

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, NLP, Computer Vision, Transfer Learning

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Web & Backend: FastAPI, Node.js, Express.js, REST APIs, React, Next.js

Databases & Tools: MongoDB, PostgreSQL, Supabase, Redis, Git, Docker, Linux, GitHub Actions, Vercel, Render

RELEVANT COURSEWORK

AI & Mathematics: Artificial Intelligence, Introduction to LLMs, Data Mining & Data Warehousing, Linear Algebra, Probability & Statistics, Operations Research

Core CS & Systems: Data Structures, Design & Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Compiler Design, Software Engineering, Automata & Formal Languages, Discrete Structures

ACHIEVEMENTS & CERTIFICATIONS

Amazon ML Summer School 2026 | *Amazon* Jul 2026 – Aug 2026

- Completed Amazon's integrated learning program on core machine learning concepts and their real-world applications.

1st Place | *Hackathon, IIIT Tiruchirapalli* 2025

- Designed and built an AI study assistant in 72 hours with conversational NLP for personalized learning.

Machine Learning Specialization | *Coursera (Stanford/DeepLearning.AI)* | credential May 2025

Crash Course on Python | *Coursera (Google)* | credential Jul 2024